Evertex® PLF-03

合成纖維抗紫外線與日光牢度增進劑

UV protection and light fastness improving agent for synthetic fibers



Everlight Chemical

Evertex® PLF-03 的應用範圍及優點

Evertex PLF-03 為聚酯（滌綸）與尼龍（錦綸）織物提供頂級的防護方案。具備卓越的紫外線吸收、反射與散射機制，能有效增進織物的耐日光堅牢度。其超群的耐高溫性能，不論是在高溫的加工製程中，或是長期處於高溫環境下，依然能展現始終如一的優異防護力，是家飾與車用紡織品的理想選擇。

特性	效益
頂級抗UV與耐候防護	• 顯著提升日光牢度與 UPF • 保持色彩鮮豔度 • 延緩纖維老化
卓越的熱穩定與防昇華性能	• 傑出的耐熱安定性 • 優異的耐昇華牢度
靈活的製程適應性與相容性	• 應用工藝廣泛 • 染料相容性絕佳 • 極佳的分散與均染性

產品特性

- 外觀：微黃色懸浮性液體
- 離子性：陰離子
- pH(100%)： 6.0 ± 1.0
- 溶解性：可以任意比例與水互溶

Characteristics

- Appearance：Pale yellowish liquid suspension
- Ionicity：Anionic
- pH(100%)：6.0 ± 1.0
- Solubility：Completely miscible with water in any ratio

應用方法

(1) 浸漬

- Evertex PLF-03可合併分散或酸性染料於高溫一浴染色
- 建議用量: 2.0% ~ 5.0% (o.w.f.)
- pH值：4.0 ~ 5.0

(2) 壓吸法

- 建議用量：20 ~ 50g/L
- 使用方法：壓吸 (壓吸率：80 ~ 100%)
- 處理條件：160 ~ 180°C ; 60 ~ 90 sec

Application

(1) Exhaustion Process (One-Bath Method)

- Evertex PLF-03 can be applied in a high-temperature one-bath process together with disperse or acid dyes.
- Dosage：2.0% ~ 5.0% (o.w.f.)
- pH Value：4.0 ~ 5.0

(2) Padding Process (Pad-Dry)

- Dosage：20 ~ 50 g/L
- Process：Padding (Pick-up：80 ~ 100%)
- Curing Conditions：160°C ~ 180°C for 60 ~ 90 seconds

Applications and Advantages of Evertex® PLF-03

Evertex PLF-03 provides premium protection solutions for polyester and nylon (polyamide) fabrics. Featuring advanced UV absorption, reflection, and scattering mechanisms, it effectively enhances the light fastness of textiles. With its exceptional thermal stability, it delivers consistent, superior protection both during high-temperature finishing processes and under long-term exposure to high-temperature environments, making it an ideal choice for home and automotive textiles.

Features	Beneifits
Premium UV Protection and Weather Resistance	• Significant Improvement in Light Fastness and UPF • Color Vibrancy Retention • Delayed Fiber Aging
Outstanding Thermal Stability and Anti-Sublimation Performance	• Excellent Heat Stability • Superb Sublimation Fastness
Flexible Process Adaptability and Compatibility	• Versatile Application Methods • Superb Dye Compatibility • Excellent Dispersion and Leveling Properties

耐日光牢度與 UPF 強化效果
Light Fastness and UPF Improvement Performance

Polyester Knits

- Disperse Orange 0.20%
- Disperse Red 0.03%
- Disperse Blue 0.36%

	20 AFU	40 AFU	UV-A cut %	UV-B cut %	UPF
Blank	3-4	3	92.93	99.7	66.7
Evertex PLF-03 2.0%	4	3-4	95.81	99.6	117.9
Evertex PLF-03 5.0%	4-5	4-5	96.82	99.7	176.7

Nylon 6 Knits 100%

- Everacid Red S-3B 0.2%

	20 AFU	40 AFU	UV-A cut %	UV-B cut %	UPF
Blank	2-3	1-2	90.01	94.21	14.2
Evertex PLF-03 2.0%	3	2	94.9	97.6	36.12
Evertex PLF-03 5.0%	3-4	2-3	95.47	98.20	48.5

備註：UPF、UVA-cut% 及 UVB-cut% 之測試結果取決於織物之組織結構、纖維種類、顏色深淺及厚薄度等多項變數。
本文件提供之數據乃本實驗室基於特定樣本測試所得，僅作為參考基準。

Notes：The test values for UPF, UVA-cut%, and UVB-cut% depend on multiple variables, including fabric structure, material type, color, and thickness.
The data provided herein is based on test results obtained from specific samples in our laboratory and is intended for reference purposes only.

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***Warnings:** The information and recommendations presented here were completed with the utmost care, but can't be extended to cover every possible case. They are intended to serve as non-binding guidelines and must be adapted to the prevailing condition. The product whether used singly or in combination with other substances, should be carried out in lab test prior to bulk production. The user needs to assume all risk and liability no matter whatsoever results are.*

bluesign SYSTEM PARTNER

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